



Software Engineering Project Presentation

AURA: Academic Unified Response & Assistance

September 2025 | Team -12

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The Challenges in Scalable Academic Support



Information Overload

Students lose track of deadlines and resources in massive PDF repositories.



Instructor Burnout

TAs & Faculty are overwhelmed by repetitive queries and manual grading.



Data Blindspots

Administrators lack data-driven insights into student confusion trends.

AURA: An Integrated AI Ecosystem



A **centralized platform** bridging the gap between students and educators using **RAG (Retrieval-Augmented Generation)** technology



Contextual AI Chatbot

Intelligent assistant providing instant answers with contextual understanding of course materials



Automated Content Creation

AI-powered generation of assessments, presentations, and study materials from course content



Real-time Analytics

Data-driven insights into student engagement, confusion patterns, and learning outcomes

Who is AURA for?



Primary Users

Instructors & TAs



Automation



Assessment Creation



Management



Secondary Users

Students



Instant Support



Learning Resources



Deadline Tracking



Tertiary Users

Administrators



Oversight



Data Analytics



Policy Development

Solving Student Challenges



"I want instant answers from course notes."



RAG-Based Chatbot

<2s latency



"I want to track upcoming deadlines."



Smart Study Assistant

Weekly Summaries



"I want personalized resource suggestions."



AI Content Recommender

Personalized learning paths



"I want queries routed to the right TA."



Auto-Routing System

Subject-based query distribution



"I want to check for duplicate questions."



Semantic Search Engine

Question similarity detection

Empowering Teaching Assistants



"I want repetitive queries auto-answered."



Auto-Response Bot

Handles common questions automatically



"I want doubts summarized from chats."



Doubt Summarizer

AI-powered discussion analysis



"I want to track unresolved queries."



Query Status Tracker

Real-time monitoring system



"I want a curated list of learning tools."



Resource Hub

Centralized teaching materials



"I want an onboarding guide."



AI Onboarding Mentor

Personalized training assistance

Tools for Instructors



"I want reports on confusing topics."

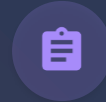


Confusion Rate Analytics

Topic difficulty visualization



"I want to auto-generate question papers."



AI Assessment Generator

Customizable test creation



"I want insights into student engagement."



Engagement Dashboard

Real-time participation metrics



"I want summarized discussion threads."



Discourse Integration

AI-powered discussion analysis



"I want to convert notes to slides."



Slide Deck Generator

AI-powered presentation creation

Administrative Oversight



User Story

"I want analytics of FAQs and unresolved queries to identify policy gaps."

- ✓ Identify knowledge gaps
- ✓ Improve response efficiency
- ✓ Data-driven policy updates



Solution



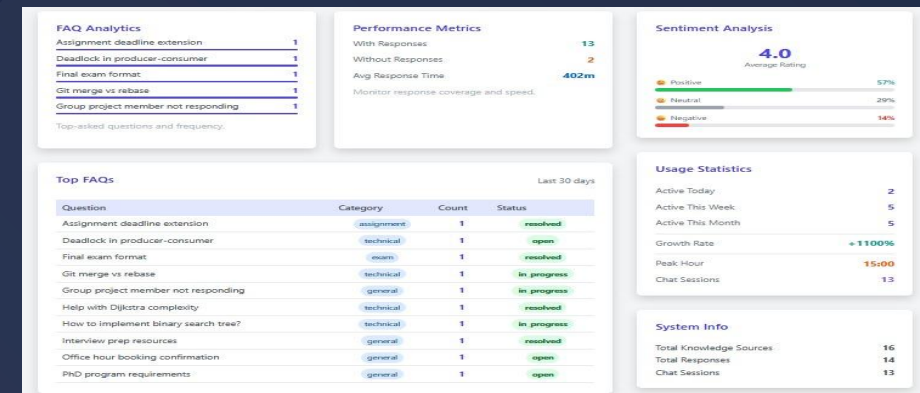
Visualizes system health



Top queries tracking



Resolution time analytics



Technical Foundation



Frontend

Vue.js (Responsive, Modular)



Backend

Python FastAPI (High performance, Async)



AI Engine

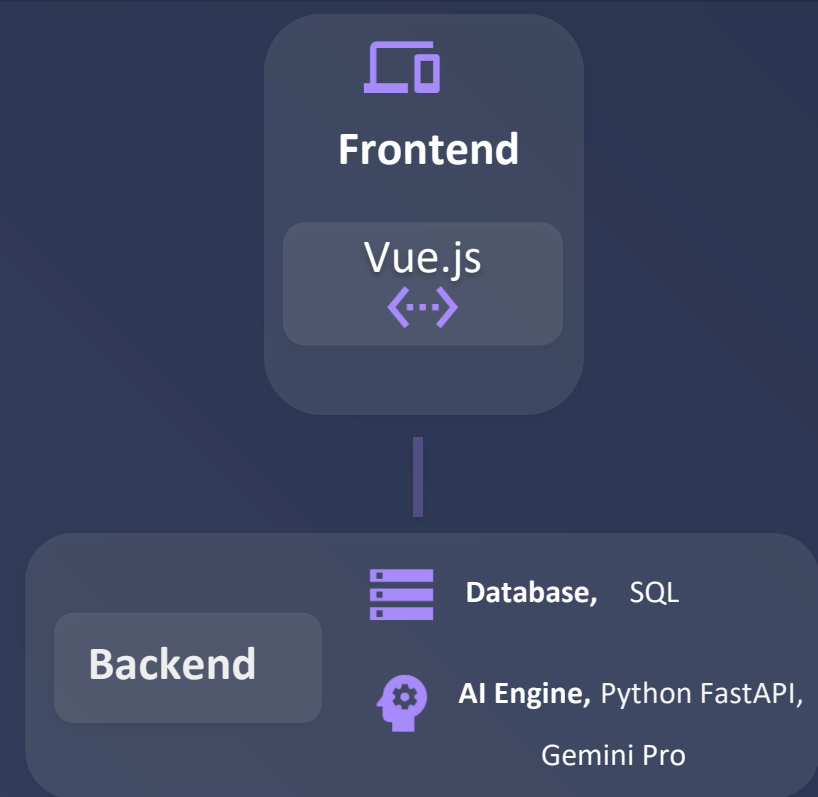
Google Gemini Pro + Vector Database (Embeddings)



Database

SQL (Relational Data Management)

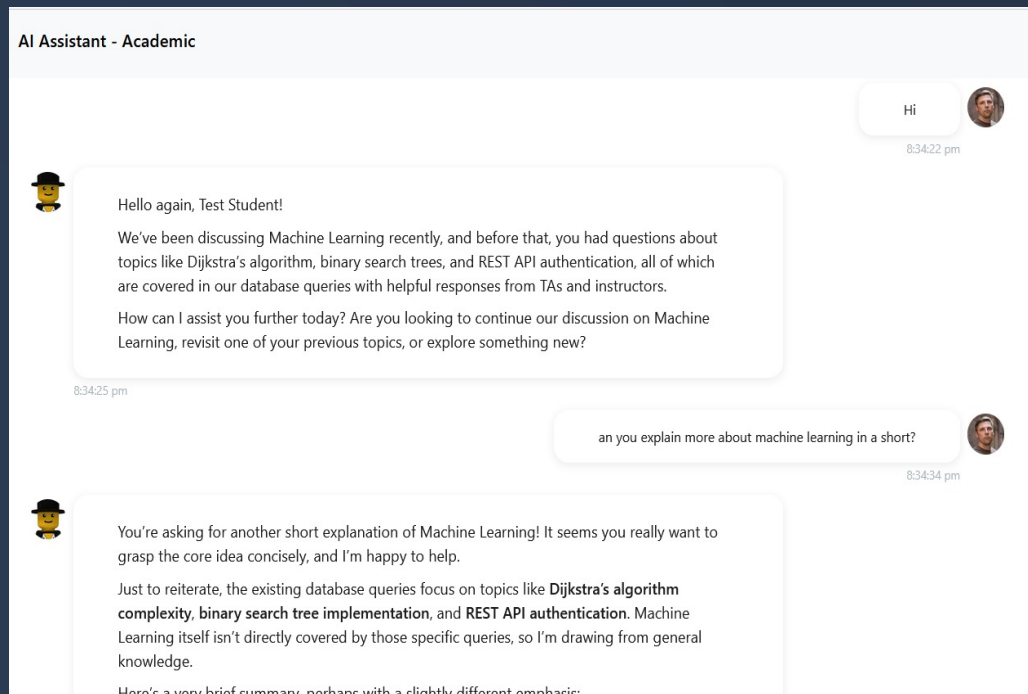
System Architecture



Core Features Live Demo



Chatbot Assistant



AI-powered chatbot providing instant answers with contextual understanding



Slide Deck Generator

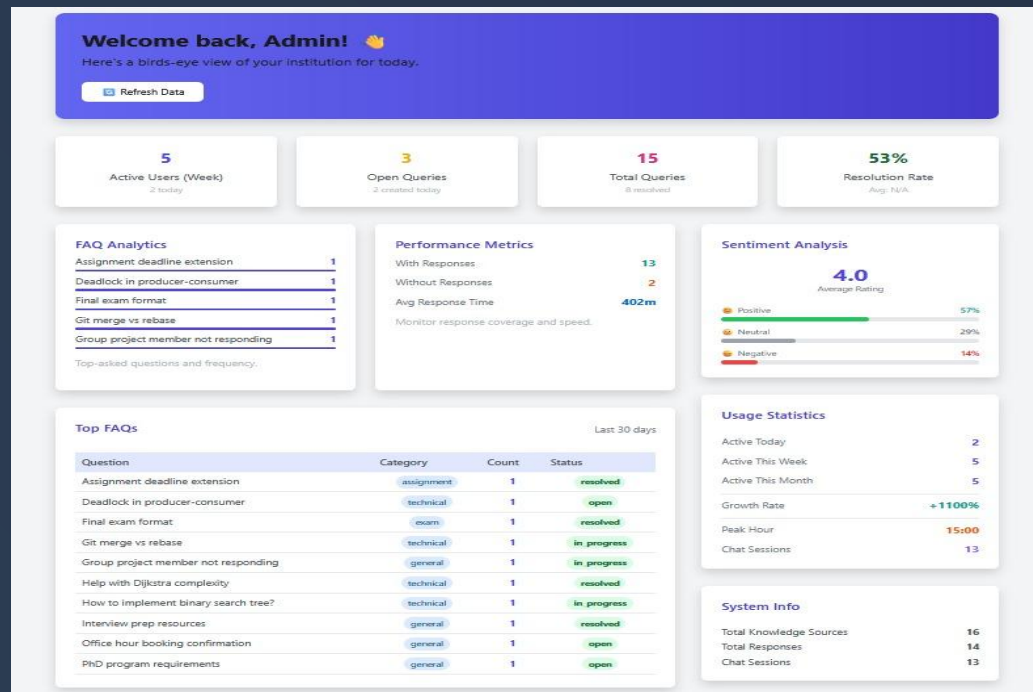
The screenshot shows a form titled "Create Slide Deck" with the subtitle "Configure your presentation settings and topics". The form includes several input fields and options: a "Course" field with a placeholder "Enter course ID", a "Presentation Title" field with a placeholder "e.g., Introduction to Machine Learning", a "Description (Optional)" field with a placeholder "Provide additional context or specific requirements...", a "Format" dropdown menu set to "Presentation (16:9)", a "Number of Slides" field set to "10", a checkbox for "Include Charts & Graphs", and a "Topics" field with a placeholder "Type a topic and press Enter or click Add" and an "Add" button. At the bottom, there is a "Back" button, a "Generate Preview" button, and a small note: "Fill all required fields: Course, Title, Topics (min 1), and 3-30 slides".

Convert lecture notes into professional slide decks with AI-powered organization

Core Features Live Demo



Engagement Dashboard



Visual analytics tracking student engagement and identifying knowledge gaps



Assessment Generator

Assessment Workspace [Reset](#) [Generate](#)

Unified AI question paper generation for Teaching Assistant

COURSE ID *
1

TITLE *
e.g. Mid-Term Logic

TOPICS *
 [Add](#)

DIFFICULTY
Medium

MARKS / QUESTION
5

TOTAL QUESTIONS
5

☐ Enable LaTeX preview

PUBLISH MODE ☒ Manual ☐ Auto

create customized assessments and quizzes tailored to learning objectives

Measuring Success



70%

Reduction in Administrative Workload



< 2s

Average Query Response Time



Deployment Status

System fully tested and ready for production

Milestone 6 Complete

Development Journey



September

Requirement Gathering

User interviews
Feature identification
Technical feasibility

Milestone 1

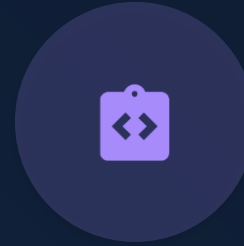


October

Wireframing & Design

UI/UX mockups
System architecture
Database schema

Milestone 2



November

Backend & AI Integration

API development
Gemini Pro integration
Vector database setup

Milestone 3-4



December

Testing & Final UI

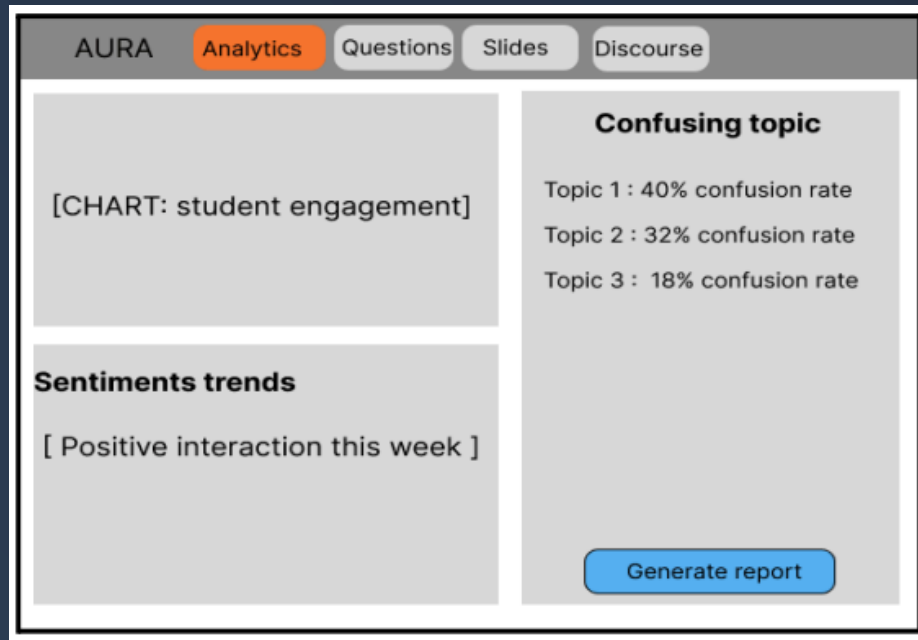
System testing
Performance optimization
Final UI polish

Milestone 5-6

From Wireframe to Final UI



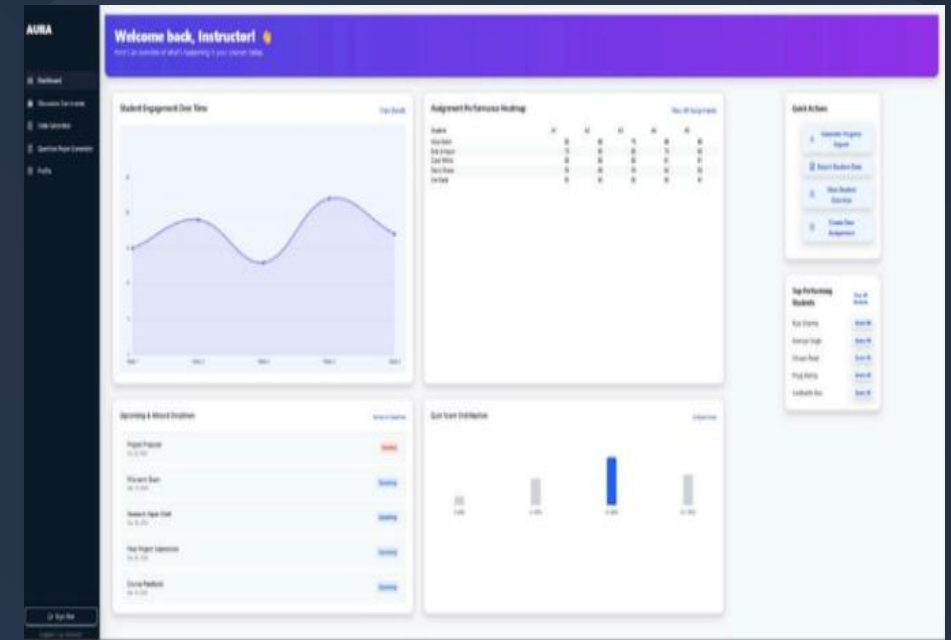
Low Fidelity Sketch



- ✓ Basic layout structure
- ✓ Component placement
- ✓ User flow mapping



High Fidelity Vue.js Screen



- ✓ Responsive design
- ✓ Interactive components
- ✓ Material styling Design

How We Worked



Process

- ✓ **Scrum Framework**
- 🕒 2-Week Sprints
- 🔄 Sprint Planning & Review
- 📉 Retrospective Sessions

Iterative Development



Tracking

- ✓ **Jira Platform**
- 📋 Kanban Boards
- ✓ Task Management
- 📊 Burndown Charts

Transparent Progress



Collaboration

- ✓ **Daily Standups**
- ⌞⌟ Git Version Control
- ⬆ Pull Request Reviews
- 💬 Slack Communication

Team Synergy

Ensuring Reliability & Security



Authentication

JWT Token Validation



Secure authentication flow with token expiration handling

Session Management



Proper session timeout and renewal mechanisms

Verified



AI Accuracy

Contextual Retention



Chatbot maintains conversation context across multiple queries

Response Accuracy



AI responses verified against educational standards

Verified



Security

Role-Based Access Control



Students blocked from Admin routes

Data Protection



User data encryption and secure storage

Verified

Milestone 6 Complete

The Road Ahead



Native Mobile App

iOS & Android applications

Offline functionality

Push notifications

Q1 2025



Voice-based Queries

Natural language processing

Multi-language support

Accessibility enhancement

Q2 2025



LMS Integration

Canvas & Moodle plugins

Single sign-on

Gradebook synchronization

Q3 2025



Our vision is to create a comprehensive educational ecosystem that seamlessly integrates with existing learning platforms while expanding accessibility and engagement.

Thank You

Team 12 - IITM BS Degree Program

